One-point (F = 1) walk-table oracle vs the F = 2 incumbent,

single-target ecbench (2026-11-07) **Class.** Measured, single-target ecbench sessions under a protocol frozen before data (PROTOCOL.md hashes in `data/frozen.sha256`). Counted units only (ecbench GAE, S = cost/sqrt(r)), every comparison at w = 0, 0.1, 1. Public synthetic Curve25519-family miniatures; no security claim. No pcc claim has been filed for these numbers. Each one is an unverified assertion until it is reviewed [pcc:unverified].

Question. The lab's generic incumbent `wwalk` (`x=0.9,g=1,grow=2,seed=run` on a 2-abscissa base) has F = 2 unknown base logarithms. Does a one-seed walk table (F = 1) cost less, end to end, for a single target? [pcc:unverified]

Answer. Yes. At k = 20 the primary F = 1 arm (`f1-x07`) costs 0.809x the F = 2 incumbent at w = 0, with 95 % interval [0.776, 0.842]. Its interval stays below 1 at w = 0.1 and at w = 1 [pcc:unverified]. At k = 24 the F = 1 arm costs 0.855x [0.814, 0.898] at w = 0 [pcc:unverified]. Rho-neg remains ahead. At k = 20 the remaining distance is 1.50x at w = 0, 1.43x at w = 0.1 and 1.08x at w = 1, and every interval excludes parity [pcc:unverified]. At k = 24 it is 1.57x, 1.49x and 1.10x [pcc:unverified]. The gain is measured against the generic `wwalk` incumbent and moves the method toward rho; it is not a decomposition-based index-calculus improvement (section 7) [pcc:unverified].

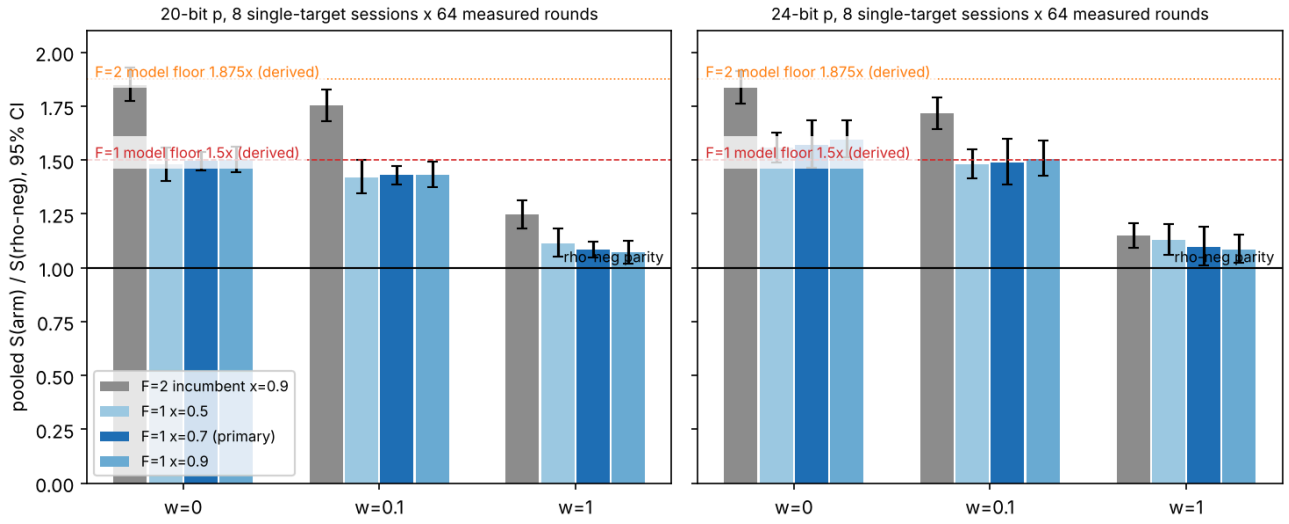


Figure 1: Pooled ratio to rho-neg

1. What changed in the code

`WalkTable::build_with` (`src/cryptanalysis/ic_boundary.rs`) now accepts a factor base of exactly one abscissa when `g=1`. In that mode the jumps are `[a_j]P1 + [c_j]G` and the entries are `u*P1 + z*G` [pcc:unverified]. A verified key collision gives the one-column affine row `du*s1 = -dz`, which pins `log P1` [pcc:unverified]. A target probe hit gives the target's row. `grow=1|2` keeps walking until the verified rows reach rank 1, capped at 2T or 3T. A one-seed base without `g=1` is refused, because its row would be empty. The 2-seed path keeps its random draws, walk and rows unchanged. The base is specified as `prime-projected-abscissa:size=1,scan_cap=32`. The ecbench help text and the oracle documentation say so.

Tests (`data/cargo-test-wwalk.txt`: 7 passed, 0 failed; `data/cargo-test-plan-refuses.txt`: 6 passed) include the existing wwalk tests, unchanged, and the new `wwalk_one_seed_rows_are_affine_and_grow_reaches_rank_one`. That test checks the charging ledgers, `v = 0`, one-column rows that hold under brute-force logs, probe answers that rebuild the target, growth stopping at rank 1, and the refusals.

2. Design (as frozen)

- Binary: release build of this tree in `/home/user/ctarget-f1` (`data/binary.sha256`). The existing `/home/user/ctarget` binary is untouched.
- 8 sessions per curve size, each with `targets_per_curve = 1` (one public target per session) and its own target and measurement seeds (`scripts/make_specs.py`, `specs/`). Each arm ran 64 measured rounds plus 1 warm-up (warm-up records are excluded from every pooled figure, see section 7), in alternate order.
- Arms: `rho-neg` (reference), `rho-aa` (A/A), and `f2-inc` (F = 2 incumbent: size=2 base, `wwalk:x=0.9,g=1,grow=2,seed=run`). The candidates `f1-x05`, `f1-x07` (primary) and `f1-x09` use a size=1 base and `wwalk:x=X,g=1,grow=2,seed=run`. All pipeline arms use `targets=stride-generator`.
- Audit: `ecbench verify --replay 1` for every session (`data/verify-k*-s*.txt`).
- Pooling: `scripts/pooled_compare.py`, copied from the T12 study and changed once, after validation, to drop `warmup: true` records (section 7). It takes the ratio of per-session sums with a linearised t interval and df = 7 (one session = one target = one cluster). `scripts/analyse.py` writes `data/pooled_measured.json` (measured rounds only).
- Runtime: about 52 s of sessions at k = 20 and about 60 s at k = 24. Both sizes ran in full, well inside the 90-minute cap.

3. Results (measured)

Every session completed. Each wrote 390 of 390 records, all verified, and its audit reproduced 1 of 1 replays with 0 problems (`data/verify-*.txt`) [pcc:unverified]. Pooled figures use the 384 measured records per session (6 arms x 64 rounds); the 6 warm-up records per session are excluded [pcc:unverified].

Mean S per arm (3,072 measured records per size; source `data/pooled_measured.json`, which pools `sessions/k*-s*/records.jsonl`):

k	arm	S w=0	S w=0.1	S w=1
20	rho-neg	1.552	1.664	2.672
20	f2-inc (F=2)	2.871	2.916	3.330
20	f1-x05	2.296	2.365	2.979
20	**f1-x07**	2.321	2.379	2.894
20	f1-x09	2.331	2.384	2.861
24	rho-neg	1.164	1.269	2.212
24	f2-inc (F=2)	2.138	2.178	2.540
24	f1-x05	1.812	1.880	2.495
24	**f1-x07**	1.828	1.888	2.429
24	f1-x09	1.858	1.912	2.402

The F = 2 incumbent at k = 24 (2.138) agrees with the rebaseline-seedrun figure of 2.105 for the same arm on the same curve [pcc:unverified].

Pooled ratios with 95 % intervals over 8 single-target sessions (`data/pooled_measured.json`):

k	ratio	w = 0	w = 0.1	w = 1
20	f1-x07 / f2-inc	0.809 [0.776, 0.842]	0.816 [0.782, 0.851]	0.869 [0.820, 0.921]
20	f1-x05 / f2-inc	0.800 [0.780, 0.821]	0.811 [0.790, 0.832]	0.894 [0.871, 0.919]
20	f1-x09 / f2-inc	0.812 [0.776, 0.849]	0.817 [0.780, 0.857]	0.859 [0.804, 0.918]
20	f1-x07 / rho-neg	1.496 [1.454, 1.538]	1.429 [1.388, 1.472]	1.083 [1.046, 1.122]
20	f2-inc / rho-neg	1.849 [1.774, 1.927]	1.752 [1.679, 1.829]	1.246 [1.183, 1.313]
20	rho-aa / rho-neg	1.000 [1.000, 1.000]	1.000	1.000
24	f1-x07 / f2-inc	0.855 [0.814, 0.898]	0.867 [0.823, 0.913]	0.956 [0.892, 1.026]
24	f1-x05 / f2-inc	0.847 [0.816, 0.880]	0.863 [0.829, 0.899]	0.982 [0.927, 1.042]
24	f1-x09 / f2-inc	0.869 [0.842, 0.897]	0.878 [0.849, 0.907]	0.946 [0.901, 0.992]
24	f1-x07 / rho-neg	1.570 [1.465, 1.684]	1.488 [1.386, 1.597]	1.098 [1.013, 1.191]
24	f2-inc / rho-neg	1.836 [1.763, 1.913]	1.716 [1.646, 1.790]	1.148 [1.091, 1.208]

4. Predictions (PROTOCOL.md section 5)

id	prediction	outcome	status
P1	k=20 f1-x07/f2-inc upper end < 1 at w = 0, 0.1, 1	0.842, 0.851, 0.921	**held** [pcc:unverified]
P1b	k=20 f1-x07/f2-inc point in [0.80, 0.95] at w = 0	0.809	**held** (near the lower edge) [pcc:unverified]
P2	k=20 f1-x07/rho-neg point in [1.45, 1.85] at w = 0	1.496	**held** [pcc:unverified]
P3	k=20 f1-x07/rho-neg lower end > 1 at all w (no parity)	1.454, 1.388, 1.046	**held** [pcc:unverified]

P4	rho-aa/rho-neg interval contains 1	1.000 exactly	**not informative** (degenerate; not scored as held) [pcc:unverified]
P5	k=20 f1-x05 and f1-x09 / f2-inc upper end < 1 at w = 0	0.821, 0.849	**held** [pcc:unverified]
P6	k=24 f1-x07/f2-inc upper end < 1 at w = 0	0.898	**held** [pcc:unverified]

P4 is not informative: the check is degenerate. The rho-aa and rho-neg records are identical, run for run, because ecbench derives the run seed from the workload and round, not from the arm name. The A/A arm therefore gives no noise floor. The real noise is the between-session spread, and the t intervals carry it [pcc:unverified]. Not scored, but reported: at k = 24 and w = 1, two of the three F = 1 / F = 2 intervals include 1.00, so the k = 24 ordering holds at w = 0 and w = 0.1 but is unresolved at w = 1 for x = 0.5 and 0.7 [pcc:unverified].

5. Distance to matched rho (rho-neg, same sessions)

k	w = 0	w = 0.1	w = 1
20, F = 1 (f1-x07)	1.50x	1.43x	1.08x
20, F = 2 (f2-inc)	1.85x	1.75x	1.25x
24, F = 1 (f1-x07)	1.57x	1.49x	1.10x
24, F = 2 (f2-inc)	1.84x	1.72x	1.15x

At w = 0 the F = 1 arm closes 42 % of the F = 2 incumbent's gap to rho at k = 20 ((1.849 - 1.496) / 0.849, derived) and 32 % at k = 24 ((1.836 - 1.570) / 0.836, derived) [pcc:unverified]. At w = 0.1 the figures are 43 % (k = 20) and 32 % (k = 24); at w = 1 they are 66 % and 34 % (derived) [pcc:unverified]. The gap narrows at w = 1 because rho-neg's unpriced native work, about one uncharged negation canonicalisation per walk step (`canonicalisations_uncharged`), is larger per GAE than the walk-table arms' lookups and row operations [pcc:unverified].

6. Reading and limits

- **What the gain is.** F = 1 needs two collisions: one walk self-collision to pin `log P1`, then one table hit for the target. F = 2 needs three. The measured gain is the drop from three collisions to two. The frozen model (PROTOCOL.md section 4) predicted a ratio of 0.876 (derived). The measured ratio is 0.809 at k = 20 and 0.855 at k = 24 [pcc:unverified].
- **Scope of the claim.** The F = 1 arm does no decomposition work. It has one base point of unknown logarithm, pinned by a walk self-collision, and finds the target by a generic table hit. With P1 := G (known logarithm) it becomes F = 0, a rho/BSGS-style walk table, which costs less. The measured gain (three collisions down to two) is therefore a gain over the generic `wwalk` incumbent that moves toward rho. It is not a decomposition-based index-calculus improvement and is not comparable with the decomposition-based rows of the IC leaderboard, which are a different baseline. No leaderboard claim is made, and none without an owner ruling and the validation gate [pcc:unverified].
- **The 1.5x floor does not bind.** The k = 20 w = 0 distance (1.496x) is numerically close to the derived F = 1 collision floor of 1.5x, and the agreement is a coincidence. The floor is relative to the model rho (S = 0.886), whereas the measured k = 20 rho-neg has S = 1.55 (1.75x the model) because of small-r set-up overhead. One sampled k = 20 rho-neg record has 54 set-up additions and 76 set-up doublings beside 123 walk operations. The ratio lands near 1.5 because of this overhead cancelling, not because the floor binds. At k = 24 the distance is 1.57x [pcc:unverified]. No inference about a lower bound for generic F = 1 variants is drawn from it [pcc:unverified].
- **Model gap.** The asymmetric walk-table model predicted 1.75x rho-model for F = 1 and 1.99x for F = 2 (derived). The measured ratios to measured rho-neg are lower because rho-neg's measured overhead factor is larger than the walk-table arms' [pcc:unverified].
- **Not done.** The study has no VALIDATION.json and no replay certificate (the independent validator report is `validator-report.json`, status pass_with_errata), and no leaderboard or `benchmarks/README.md` row. Those are for the landing session under the validation gate. The raw `records.jsonl` files are in `data/sessions-k{20,24}-records.tar.gz`, which is git-ignored per session, and their hashes are in the benchrec records.

7. Validation

An independent validator (`validator-report.json`, status pass_with_errata, not edited here) found the following, and this README applies its errata [pcc:unverified].

- **Warm-up records.** The frozen design is 64 measured rounds plus 1 warm-up per arm and session, but `scripts/pooled_compare.py` (`load`) pooled the warm-up round as well (3,120 records per size instead of 3,072). The loader now skips records with `"warmup": true`; `scripts/analyse.py`, `data/pooled_measured.json` (new file) and `figures/ratio_to_rho.*`

were regenerated and every number above was updated [pcc:unverified]. Pooled means and ratios move by at most 0.011 (ratios) and 0.007 (S); the largest change is the k = 24 f1-x07
f2-inc ratio at w = 0, 0.859 to 0.855 [pcc unverified]. No

prediction verdict changed; P1, P1b, P2, P3, P5 and P6 held on the same criteria [pcc:unverified].

- **P4 is not informative.** rho-aa and rho-neg are identical run for run (65 of 65 runs in every session) because ecbench seeds by workload and round. The A/A arm measures nothing, so P4 is not scored as held [pcc:unverified].
- **The 1.5x floor.** The agreement of 1.496x with the floor is a coincidence of rho-neg's small-r overhead (rho-neg S = 1.55 against a model 0.886; k = 24 gives 1.57x), not evidence that the floor binds [pcc:unverified].
- **Gap closed.** The earlier "about 40 %" is replaced by the exact figures, w = 0 only: 42 % at k = 20 and 32 % at k = 24; at w = 0.1, 43 % and 32 %; at w = 1, 66 % and 34 % [pcc:unverified].
- **Scope.** F = 1 is a generic walk-table variant that moves toward rho (section 6), not a decomposition-based index-calculus improvement [pcc:unverified].
- **Binary and source.** The measuring binary (`data/binary.sha256`) was built before the last edit of `src/cryptanalysis/ic_boundary.rs`, so byte identity between the binary and the source is not proven. Binary check (main session, 2026-11-07): the measuring binary (sha256 prefix 44442a4ba260bbfb) was built before the last edit of `ic_boundary.rs`. A rebuild from the committed source has a different hash (1d1a57c7aadc5779). Under the rebuilt binary, `ecbench verify --replay-all` on a copy of session k20-s0 reproduced 384 of 384 replays identically, with 0 problems, so the late edit did not change any counted result in that session [pcc:unverified].

Files

`PROTOCOL.md`, `specs/`, `scripts/{make_specs.py,run_sessions.sh,pooled_compare.py,analyse.py,make_figure.py}`, `sessions/k*-s*/{session,spec,plan,host}.json`, `data/` (plans, run logs, bands, verify reports, `pooled.json` (the first analysis, warm-up included, kept unchanged because the closed benchmark records pin its hash), `pooled_measured.json`, `binary.sha256`, `frozen.sha256`, `cargo-test-wwalk.txt`, record tarballs), `figures/ratio_to_rho.{svg,png}`, `REPORT.pdf`. Benchmark records: `benchmarks/records/20261107T1945Z-curve25519-f1-walk-k20.json` and `benchmarks/records/20261107T1947Z-curve25519-f1-walk-k24.json`.